MENZAL TECHNOLOGIES · TWIN AI POC

Urban Thermal Intelligence from Earth Observation Data

A visual walkthrough of Twin AI PoC. From Sentinel-2 surface features and unsupervised clustering to Landsat LST prediction.

Sentinel-2C Level-2A

Landsat-9 Collection 2 Level-2

Random Forest Regression

KMean Clustering

01 · INPUTS & STUDY CONTEXT

Two EO products, one analysis context

Sentinel-2C · Optical surface information

Level-2A surface reflectance. Bands B02, B03, B04, B08, B11 and B12 feed spectral indices and clustering.

Landsat-9 · Thermal reference

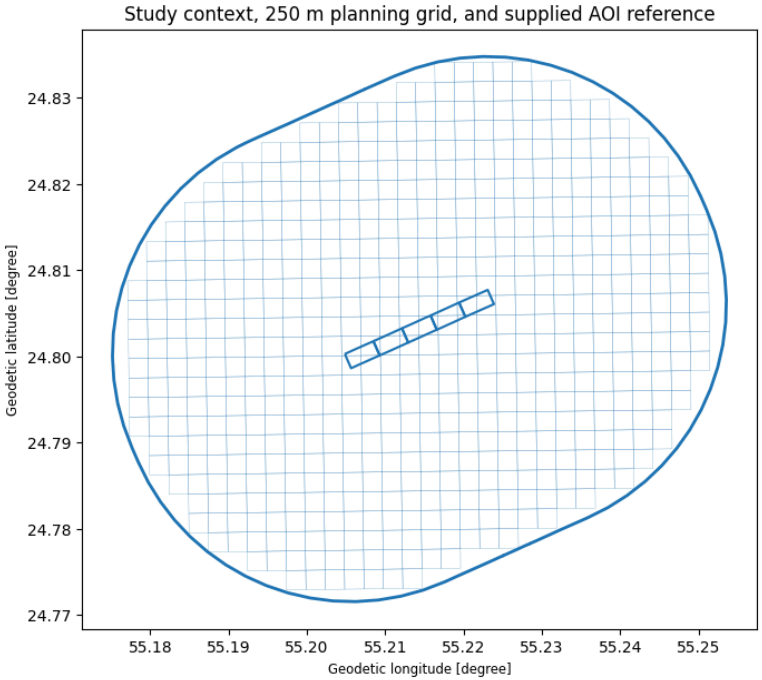
Collection 2 Level-2 Science Product. ST_B10 supplies the observed land-surface temperature target.

612

250 m planning cells generated across the broader study context.

Study setup

The supplied five AOI polygons are retained as geographic reference; modeling is performed over a 3 km buffered context.



Study context, regular 250 m grid, and the original supplied AOI segments.

Sentinel-2C
10/20 m optical reflectance → **Feature engineering**
indices + surface regimes → **Landsat-9**
30 m LST target grid → **Urban indicators**
250 m planning layer

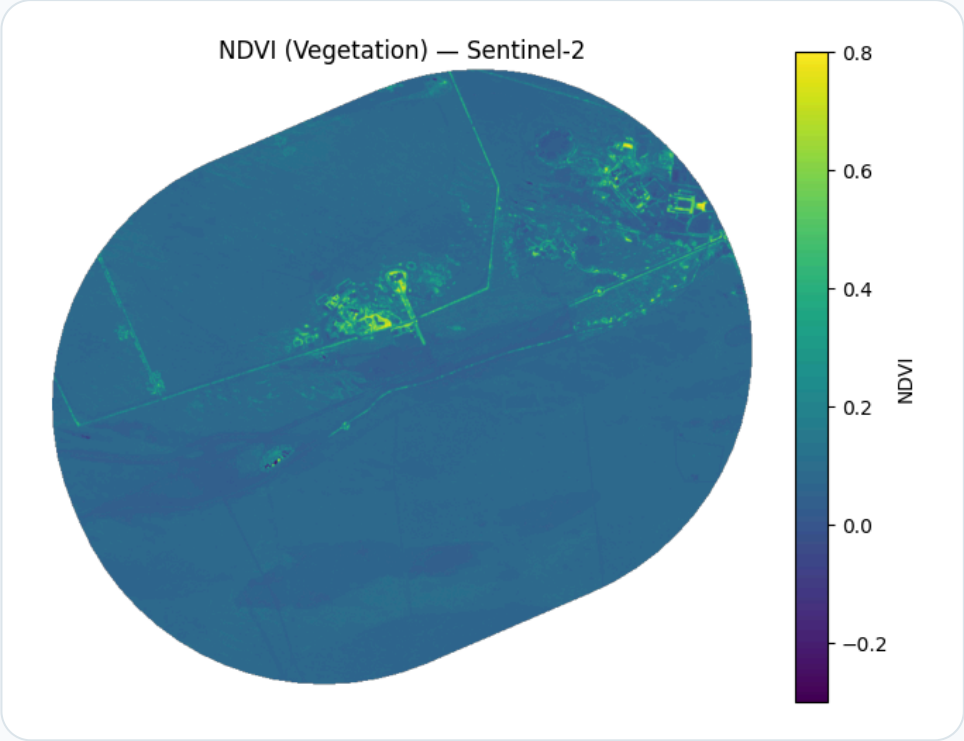
From optical imagery to vegetation signal

Input: true-colour Sentinel-2 view



The optical scene provides the surface context used for spectral features.

Output: NDVI



NDVI highlights vegetation-like responses using the NIR and red bands.

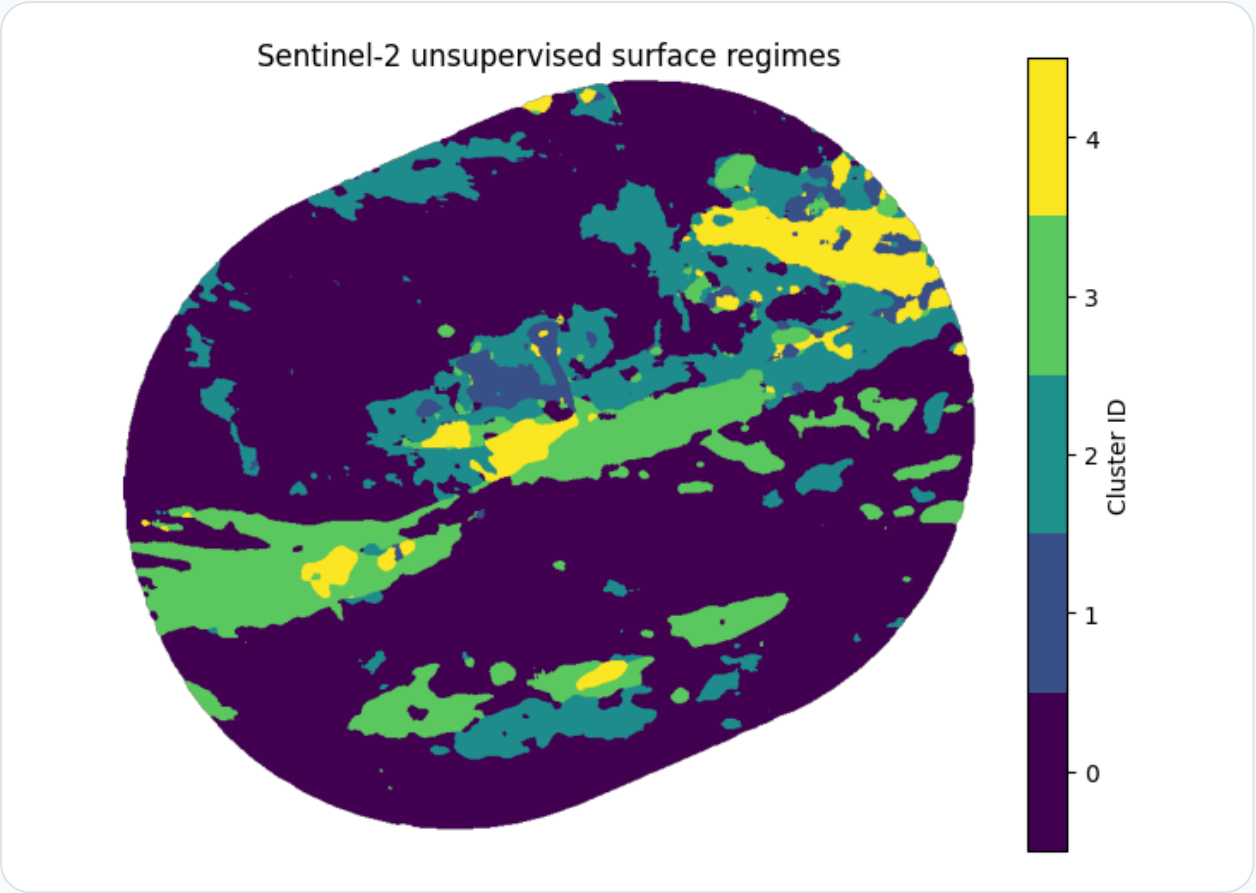
```
NDVI = (B08 - B04) / (B08 + B04)
```

The notebook also derives NDBI, MNDWI and BSI. Note that these indices are used as spectral features, not as definitive semantic land-cover labels.

419,510 valid Sentinel-2 pixels | **10 m analysis grid** | **Cloud/shadow masking via SCL**

03 · UNSUPERVISED SURFACE-REGIME CLUSTERING

Discovering recurring spectral surface patterns



Model setup

10 spectral features → StandardScaler → MiniBatchKMeans

5
CLUSTERS

200k
FIT PIXELS

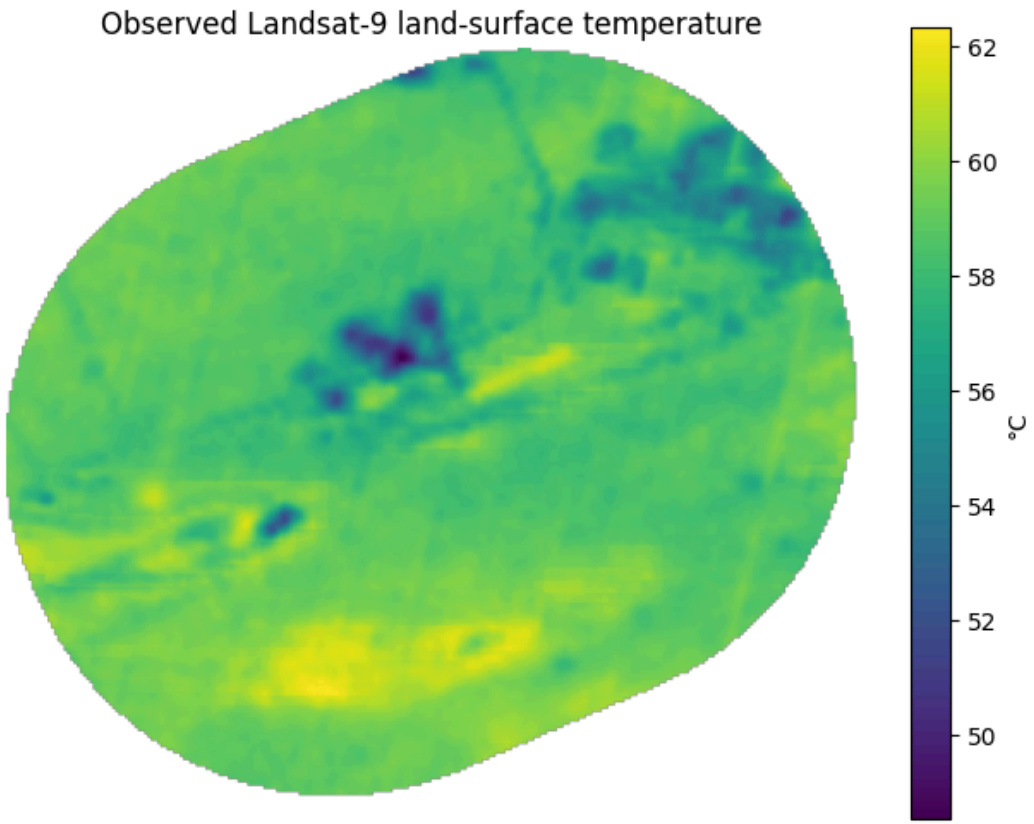
Notebook interpretation

- **Cluster 1** is the most vegetation-like regime.
- **Cluster 4** is the most water-like regime.
- **Cluster 3** is later found to have the highest mean observed LST.

No independent land-cover labels are available in the data. The clusters are therefore reported as **possible surface regimes**, not validated land-cover classes.

04 · LANDSAT THERMAL OBSERVATION

Observed land-surface temperature



48.53°C
MINIMUM

58.81°C
MEDIAN

62.33°C
MAXIMUM

46,616
VALID LST CELLS

Processing in the notebook

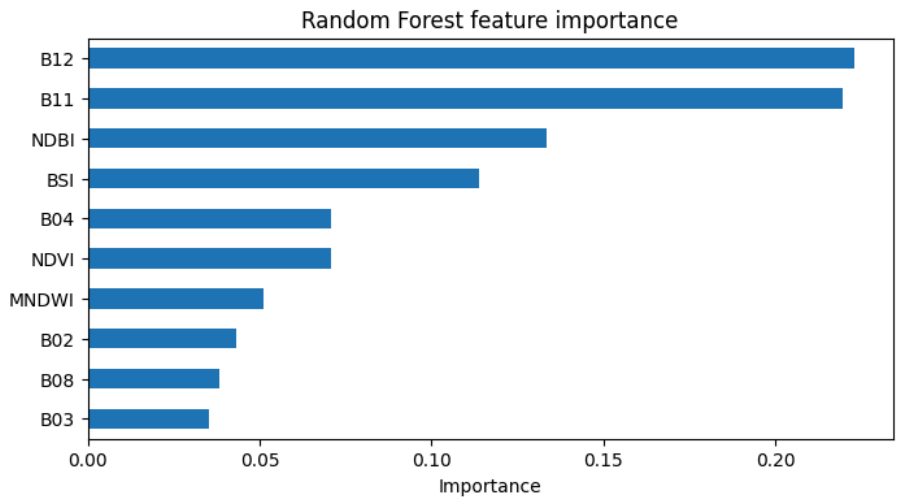
- Use Landsat Level-2 **ST_B10**.
- Apply QA masking for invalid/cloud/saturated observations.
- Apply the scene metadata scale and offset.
- Convert the final temperature values to °C.

ST_B10 is already the Level-2 LST product. We did not rederive LST from raw thermal radiance.

05 · RANDOM FOREST LST REGRESSION

Can Sentinel-2 surface features reproduce the Landsat thermal pattern?

0.512°C	0.713°C	0.532	-0.065°C
MAE	RMSE	R^2	BIAS (PRED - OBS)



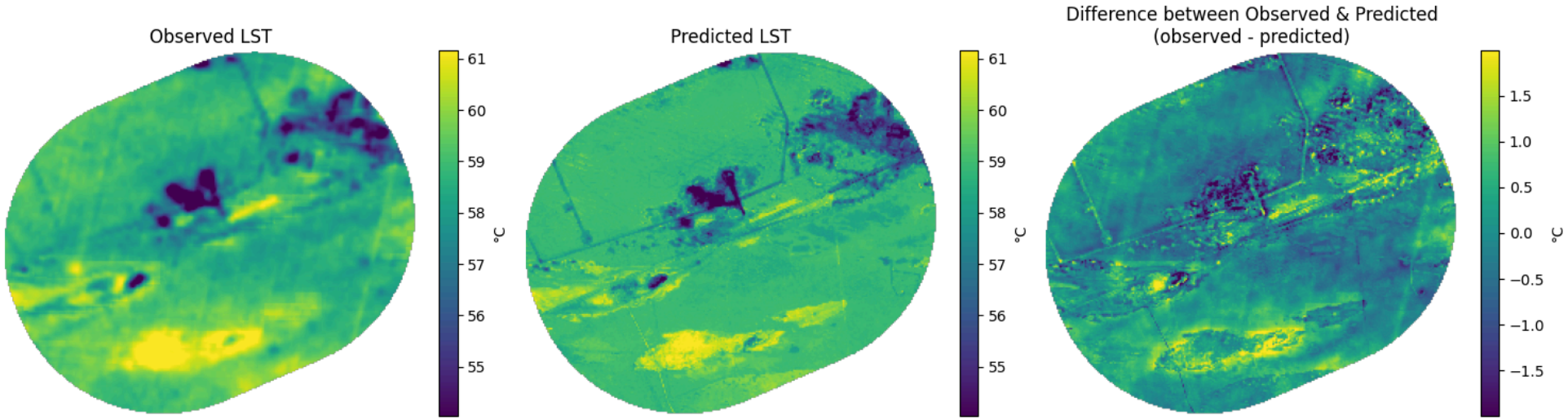
Feature importance from the fitted Random Forest. SWIR bands B12 and B11 are the strongest individual predictors in this run.

Spatial validation setup

ITEM	VALUE
Total usable samples	46,569
Train / test samples	34,899 / 11,670
Spatial block size	300 m
Train / test groups	383 / 128
Split	GroupShuffleSplit (~75/25)

06 · PREDICTION & RESIDUAL ANALYSIS

Observed vs predicted LST



Observed

QA-masked Landsat ST_B10 in °C.

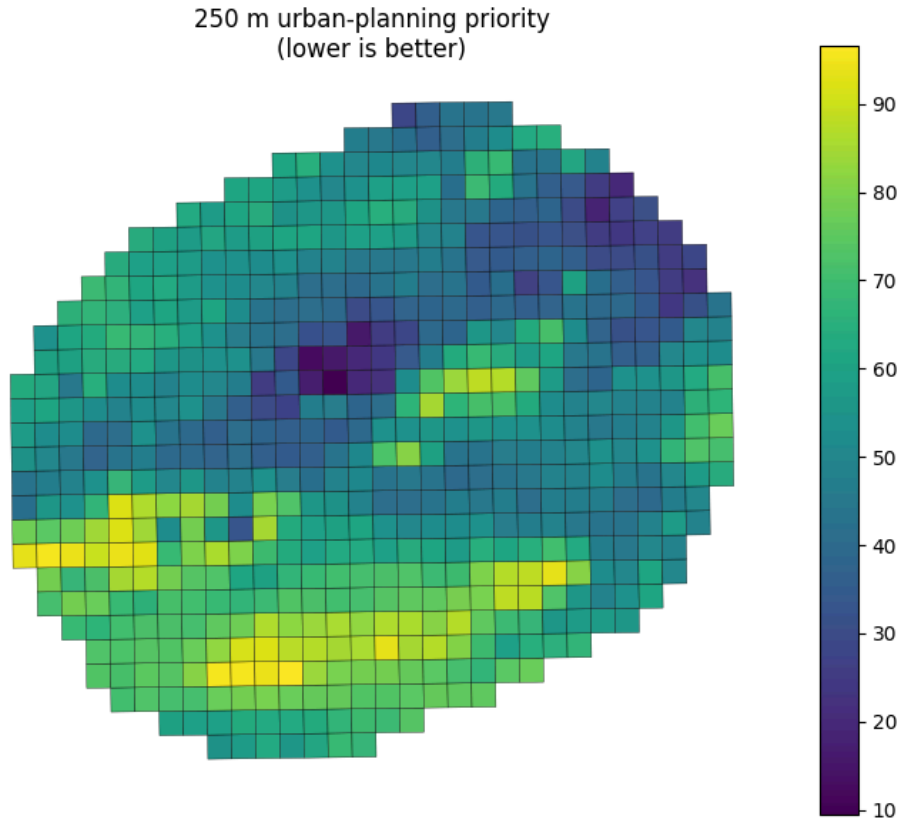
Predicted

Random Forest prediction from aggregated Sentinel-2 features.

Residual

Observed - predicted. Positive values are hotter than the model expected.

Translate EO and model outputs into comparable planning cells



Priority = 0.50·Thermal Pressure + 0.30·Vegetation Deficit + 0.20·Heat-Associated Surface Fraction

Example high-priority cell · G004_012

96.58 PRIORITY SCORE	**61.80°C** MEAN LST
99.82% THERMAL PRESSURE	**83.36%** HEAT-ASSOCIATED SURFACE

What each cell can answer

- How hot is the area?
- How much is vegetation-like?
- Which surface regime dominates?
- How unusual is its thermal pressure?
- How well does the LST model explain it?

A reproducible EO → AI → planning-intelligence pathway

EO inputs
Sentinel-2C + Landsat-9

→

Preprocessing
masking, scaling, alignment

→

Features
bands + NDVI/NDBI/MNDWI/BSI

→

AI
clustering + RF regression

→

Planning layer
250 m indicators

Primary notebook outputs

NDVI
Vegetation-sensitive surface feature.

Surface regimes
Unsupervised spectral clustering.

Observed LST
Landsat thermal reference layer.

Predicted LST
Sentinel-2 → LST model output.

Residuals
Observed - predicted differences.

Planning priority
250 m interpretable ranking.

Interpretation boundaries

- gIQ is the delivery/access platform; source-product identity comes from the supplied Sentinel-2 and Landsat metadata.
- The supplied Sentinel-1 scene is excluded because it does not overlap the UAE pilot.
- One observation period does not support urban-growth or change claims.
- Clusters are surface regimes, not independently validated land-cover classes.
- LST is surface-skin temperature, not air temperature or human heat exposure.
- The RF metrics validate prediction against withheld Landsat LST cells, not independent ground truth.

The value of the PoC is the reproducible pathway from EO data to explainable planning indicators, with observed data, model predictions, and derived scores kept separate.